Preliminary Progress Review (PPR) Report of 25_26J_009

Project ID: 25_26J_009

Project Title: A Lightweight Trust Management Framework in the Networking Layer of IoT Devices.

Title of the Individual Component: Distributed Isolation, Mitigation, and Recovery Framework for RPL-based IoT Networks

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1. Introduction

1.1. Purpose

This Preliminary Progress Review (PPR) document presents the current state of progress for the individual research component focused on Distributed Isolation, Mitigation, and Recovery mechanisms within the broader project titled "A Lightweight Trust Management Framework in the Networking Layer of IoT Devices." The document outlines the research pathway and justifies the credibility and viability of the proposed component in accordance with the module requirements for final-year research projects.

This component specifically addresses the post-detection phase of security incidents in RPL-based IoT networks, a critical yet often overlooked aspect of trust-aware routing security. The PPR describes the research problem, identifies gaps in existing literature, states the research questions and objectives, proposes a methodology, and documents progress made thus far in the project lifecycle.

The intended audience includes supervisors, co-supervisors, panel evaluators, and academic reviewers responsible for assessing the progress of the research project. The document is structured to facilitate evaluation by presenting completed work in a systematic and evidence-supported manner, while also outlining the strategy for upcoming research phases.

1.2 Definitions, Acronyms, and Abbreviations

Acronym / Term	Definition / Description
IoT	Internet of Things
RPL	IPv6 Routing Protocol for Low-Power and Lossy Networks
LLN	Low-Power and Lossy Network
DODAG	Destination-Oriented Directed Acyclic Graph
6LoWPAN	IPv6 over Low-Power Wireless Personal Area Networks
IDS	Intrusion Detection System
DIO	DODAG Information Object
DAO	Destination Advertisement Object
DIS	DODAG Information Solicitation
PDR	Packet Delivery Ratio
PPR	Preliminary Progress Review
WBS	Work Breakdown Structure
Cooja	Network simulator used for Contiki-NG
Contiki-NG	Open-source operating system for constrained IoT devices
Sinkhole Attack	Attack where a malicious node falsifies routing metrics to attract traffic
Selective Forwarding	Attack where a node drops selected packets while forwarding others

Blacklist	List of confirmed malicious nodes excluded from routing
DODAG Healing	RPL mechanism for reconstructing routing topology after disruption
Trust Score	Quantitative measure of node reliability derived from behavioral metrics

1.3 Overview

The remainder of this PPR is organized as follows. Section 2 presents a comprehensive literature review that establishes the state of the art in RPL security, trust management, and post-detection resilience mechanisms. Section 2.1 identifies and justifies the significance of the research problem. Sections 3 and 4 state the research questions and objectives, respectively, defining the scope and expected contributions of the individual component.

Section 5 describes the research methodology including the simulation-based experimental approach, system architecture, and evaluation framework. Section 6 details the sources of test data and analysis techniques. Sections 7 and 8 outline the anticipated benefits and the scope of deliverables. Section 9 discusses research constraints and limitations. Section 10 presents the project schedule, followed by references and appendices.

2. Literature Review

Securing RPL-based IoT networks against routing-layer attacks has received considerable research attention, yet the post-detection phase — comprising node isolation, attack mitigation, and network recovery — remains significantly underexplored. While numerous studies have addressed attack detection and trust computation, the mechanisms required to act upon detection outcomes and restore network integrity are largely absent from the literature. This review examines existing work on RPL resilience, isolation frameworks, mitigation strategies, and recovery protocols to identify critical gaps that this research addresses.

Post-Detection Resilience in RPL Networks

The majority of existing RPL security research terminates at the point of attack detection, with limited attention paid to what occurs after a malicious node has been identified. Raoof, Matrawy, and Lung (2019) provide a comprehensive survey of RPL routing attacks and mitigation methods, noting that while detection mechanisms have matured, corresponding response and recovery procedures are not standardized and often left to implementation-specific decisions. This creates a critical vulnerability window between detection and remediation, during which adversarial nodes continue to influence routing behavior and degrade network performance.

Verma and Ranga (2020) similarly identify the absence of holistic post-detection frameworks as a major gap in RPL security, arguing that detection without effective isolation and recovery mechanisms provides incomplete protection. Their review of RPL security measures across IEEE Sensors Journal highlights that most proposed solutions either focus exclusively on detection accuracy or propose isolated mitigation steps without considering the broader operational continuity of the network.

Node Isolation Mechanisms

Several studies have proposed lightweight approaches to excluding malicious nodes from RPL routing. Al-Sarawi et al. (2023) developed a passive rule-based sinkhole detection mechanism that flags anomalous nodes but does not provide a distributed consensus mechanism for acting upon those flags. This illustrates a recurring limitation: isolation decisions are made locally without coordination, leading to inconsistent blacklisting across the network and potential for legitimate nodes to be incorrectly excluded based on incomplete neighbor-level information.

The SMTrust protocol by Muzammal et al. (2020) incorporates trust-score-based routing decisions that effectively avoid low-trust nodes; however, this approach conflates avoidance with isolation. Nodes are not formally removed from the network topology — they simply receive fewer routing requests. As a result, malicious nodes continue to broadcast DIO messages and participate in topology formation, allowing them to re-attract traffic once detection confidence decreases. A formal distributed isolation protocol that achieves consensus-based node removal and maintains consistent blacklists across the DODAG is absent from this work.

Khan et al. (2025), in their FL-DSFA framework using federated learning for selective forwarding attack mitigation, demonstrate that collaborative detection significantly improves isolation accuracy. However, the federated learning overhead is prohibitive for severely resource-constrained IoT motes (typically 8–32 KB RAM), and the isolation process still lacks a secure re-authentication mechanism to prevent isolated nodes from rejoining the network through identity spoofing.

Attack Mitigation Strategies

Attack mitigation in RPL literature broadly encompasses traffic rerouting, quarantine mechanisms, and emergency communication channels. Patel and Shah (2020) experimentally evaluate RPL performance under simultaneous sinkhole and selective forwarding attacks,

demonstrating that standard RPL's reactive rerouting via DODAG reformation introduces significant delays of 30–90 seconds before stable alternative paths are established. This delay is unacceptable for critical IoT applications such as healthcare monitoring and industrial control systems where sub-10-second response times are required.

Hashemi and Aliee (2020) propose a fuzzy, dynamic, and trust-based routing protocol that dynamically adjusts routing preferences based on trust levels, providing a form of soft mitigation by reducing reliance on suspicious nodes. However, their approach does not implement hard traffic redirection or quarantine states for nodes under investigation, leaving the network vulnerable during the trust-evaluation window. Adarbah et al. (2022) address the specific challenge of selective forwarding mitigation through an ECDH-based authentication protocol, but this cryptographic approach introduces significant computational overhead that conflicts with the lightweight design requirements of constrained IoT devices.

Krari et al. (2025) present a multi-attack detection approach using deep learning for RPL routing threats, demonstrating high detection accuracy but acknowledging that the computational demands of deep learning models are incompatible with resource-constrained environments. Their work does not address the mitigation phase following detection, reinforcing the pattern identified across the literature where detection and mitigation are treated as independent, disconnected concerns.

Network Recovery and Topology Reconstruction

Network recovery following malicious node removal is one of the least studied aspects of RPL security. The standard RPL healing mechanism described in RFC 6550 relies on DODAG version incrementation to trigger global topology reconstruction, which incurs substantial control message overhead and convergence delays. Yadav and Kaur (2025) demonstrate that naive DODAG version incrementation in response to rank attacks can cause cascading topology instability, particularly in networks with high node density or significant topological dependence on the isolated node.

Poornima, Vimala, and Shreyas (2025) propose a cross-layer RPL objective function that improves energy-aware routing recovery; however, this work does not specifically address the scenario where topology disruption arises from malicious node isolation rather than energy depletion. The distinction is significant: energy-based recovery can proceed gradually, whereas security-triggered recovery must balance speed with stability to prevent the network from oscillating between multiple DODAG configurations.

Mokrani et al. (2024) introduce LEA-RPL, a lightweight energy-aware variant of RPL that uses particle swarm optimization for route selection. While this approach provides improved convergence properties, it does not address secure node re-entry following isolation or the prevention of previously blacklisted nodes from exploiting topology reconstruction events to re-infiltrate the network. Alhandi et al. (2025) propose a lightweight trust evaluation model that includes mechanisms for progressive trust rehabilitation, representing the closest existing work to a re-authentication framework, but their evaluation focuses on trust computation rather than the security of the re-entry process itself.

Integration and Interoperability Challenges

A consistent finding across the literature is the absence of integrated frameworks that connect detection outputs directly to isolation, mitigation, and recovery actions in a coordinated pipeline. Prajapati et al. (2025), in a comprehensive survey of RPL routing attacks and defences, identify the lack of end-to-end security management systems as a primary barrier to practical deployment. Most proposed solutions require modifications to the RPL protocol stack that are incompatible with RFC 6550 specifications, creating interoperability challenges with legacy IoT infrastructure.

Furthermore, existing frameworks typically evaluate security and performance metrics in isolation rather than assessing the trade-offs between security enhancement, energy consumption, memory utilization, and convergence time under realistic combined attack scenarios.

2.1. Identification and Significance of the Problem

Despite significant advances in RPL attack detection and trust-based routing, the field has not addressed the critical operational challenge that follows detection: how to safely, efficiently, and verifiably remove malicious nodes from the network, limit the damage of ongoing attacks during the removal process, reconstruct the routing topology, and prevent re-infiltration. This post-detection management gap represents a fundamental barrier to deploying trust-aware RPL security mechanisms in real-world IoT environments.

Current solutions suffer from three compounding deficiencies. First, isolation decisions are made unilaterally at individual nodes without distributed consensus, resulting in inconsistent blacklists and false isolation of legitimate nodes experiencing temporary link degradation. This is particularly problematic in RPL networks where topology is dynamic and node behavior naturally varies under lossy channel conditions. Second, the transition period between attack detection and successful node isolation is currently unmanaged: no mechanisms exist to redirect traffic away from suspected nodes, maintain emergency communication paths, or ensure critical IoT services remain operational during this window.

Third, existing recovery mechanisms rely on standard RPL DODAG reformation, which is neither optimized for the security-triggered isolation scenario nor designed to prevent re-entry by previously isolated nodes. The absence of a secure re-authentication framework means that isolated malicious nodes can rejoin the network after the topology has stabilized, re-establishing their position through identity spoofing or gradual trust rehabilitation exploitation.

The significance of addressing this gap extends beyond academic interest. IoT deployments in smart cities, healthcare monitoring, and industrial automation rely on RPL networks for continuous, reliable data delivery. The inability to respond effectively to confirmed routing attacks undermines the security value of detection mechanisms and leaves operational networks vulnerable to prolonged service disruption. This research directly addresses these deficiencies by proposing an integrated, lightweight, and standards-compatible framework for distributed isolation, attack mitigation, and secure network recovery in RPL-based IoT environments.

3. Research Questions

The following research questions guide the design, implementation, and evaluation of the proposed distributed isolation, mitigation, and recovery framework for RPL-based IoT networks:

RQ1: How can a distributed consensus mechanism be designed to achieve coordinated and consistent malicious node isolation across all participating nodes in an RPL DODAG, without requiring centralized control or significant additional communication overhead?

RQ2: What lightweight traffic rerouting and quarantine mechanisms can minimize attack impact and maintain critical IoT service availability during the interval between attack detection and the completion of node isolation?

RQ3: How can topology reconstruction algorithms be designed to restore optimal RPL routing performance within 30 seconds following malicious node isolation, while preserving network partition avoidance and load balance?

RQ4: What multi-factor authentication and progressive trust rehabilitation mechanisms are sufficient to prevent previously isolated malicious nodes from rejoining the network through identity spoofing or trust score manipulation?

RQ5: How can isolation, mitigation, and recovery processes be designed to operate within the resource constraints of IoT devices (8–32 KB RAM, constrained CPU) while maintaining effectiveness against combined sinkhole and selective forwarding attack scenarios?

4. Research Objectives

The primary goal of this individual research component is to design, implement, and evaluate a lightweight, distributed framework for post-detection security management in RPL-based IoT networks — encompassing coordinated node isolation, attack impact mitigation, and secure network recovery — while maintaining compatibility with standard RPL implementations and operating within the resource constraints of IoT devices.

This overarching goal is decomposed into the following specific sub-objectives:

Sub-Objective 1: Design a Distributed Malicious Node Isolation Framework

To develop a distributed isolation protocol that enables coordinated removal of confirmed malicious nodes from active routing participation. The framework must implement a lightweight consensus mechanism for isolation decisions, maintain consistent blacklists across network nodes, prevent false isolation of legitimate nodes, and ensure isolation decisions are executed within 2–3 seconds of receiving attack confirmation from the detection module.

Sub-Objective 2: Implement Lightweight Attack Impact Mitigation Strategies

To create immediate mitigation mechanisms that limit attack damage while node isolation is being processed. This includes developing traffic redirection protocols to route packets away from suspected nodes, implementing temporary quarantine states for nodes under investigation, creating emergency communication channels that bypass compromised routing paths, and ensuring critical IoT services remain operational during attack response with less than 10% throughput reduction.

Sub-Objective 3: Develop Network Recovery and Topology Reconstruction Algorithms

To design comprehensive recovery mechanisms that restore optimal network functionality after malicious node isolation. This includes implementing rapid alternative route discovery for affected network segments, developing topology healing algorithms that reconnect isolated network partitions, creating load balancing mechanisms to redistribute traffic after topology changes, and ensuring recovery processes complete within 30 seconds to minimize service disruption.

Sub-Objective 4: Establish a Secure Re-authentication and Trust Rehabilitation Framework

To implement robust mechanisms that prevent isolated malicious nodes from rejoining the network through identity spoofing or exploitation of trust rehabilitation processes. This includes developing multi-factor node authentication protocols, implementing progressive trust rebuilding procedures for nodes seeking network re-entry, creating secure node identity verification mechanisms, and establishing enhanced monitoring periods for rehabilitated nodes.

Sub-Objective 5: Validate Integration with Detection and Trust Systems

To demonstrate seamless integration of the isolation, mitigation, and recovery mechanisms with the attack detection and trust computation modules developed by collaborating team members. This includes implementing standardized interfaces for receiving attack notifications, validating coordinated response with dynamic parent selection mechanisms, and conducting comprehensive testing using Contiki-NG and the Cooja simulator.

5. Research Methodology

This research employs a simulation-based experimental approach to design, implement, and validate the proposed distributed isolation, mitigation, and recovery framework for RPL-based IoT networks. This methodology is consistent with the broader project strategy and with established practices in RPL security research, where the Contiki-NG operating system and Cooja network simulator provide a reproducible, cost-effective, and controllable evaluation environment suitable for iterative development of lightweight security mechanisms.

The methodology is structured into four phases: baseline and attack scenario modeling, framework design and implementation, integration and system testing, and performance and security evaluation.

Phase 1: Baseline and Attack Scenario Modeling

RPL-based IoT network topologies are constructed in Cooja using Contiki-NG, configured with standard RPL (RFC 6550) and varying node counts of 20, 50, and 100 nodes. Baseline simulations are run under normal (attack-free) conditions to establish performance benchmarks for key metrics including Packet Delivery Ratio (PDR), DODAG formation time, control message overhead, and node energy consumption. These baselines inform the acceptable overhead thresholds for the proposed framework.

Attack scenarios are then introduced, beginning with isolated sinkhole attacks, isolated selective forwarding attacks, and finally combined simultaneous attacks. The behavior of the RPL topology under each attack type is characterized to understand the specific isolation and recovery challenges posed by each scenario. Particular attention is paid to the topology dependence on compromised nodes — nodes whose removal would partition the network or create orphaned subtrees — as this directly informs the design of recovery algorithms.

Phase 2: Framework Design and Implementation

The isolation, mitigation, and recovery framework is designed as a distributed module that operates at the RPL routing layer on each individual node, consistent with the lightweight, decentralized design philosophy of the broader project. The framework receives attack alerts from the IDS component developed by a collaborating team member and produces routing decisions that are consumed by the trust-aware parent selection module.

Isolation Module Design

Upon receiving an attack alert with an identified malicious node ID, the isolation module initiates a lightweight distributed consensus process. Each node receiving the alert broadcasts a signed isolation vote using RPL DIO message extensions. When a threshold number of neighboring nodes confirms the alert (configurable based on network density), the malicious node is added to the local blacklist and excluded from parent selection and routing path calculations. Blacklist entries are propagated through the DODAG via modified DAO messages to ensure network-wide consistency.

To prevent false isolation due to transient link failures, the module implements a confirmation delay mechanism: nodes receiving a single unconfirmed alert enter a monitoring state rather than immediately isolating the suspect node. Isolation is only executed when the alert is corroborated by observed local behavioral anomalies or confirmed by a quorum of neighboring nodes within a configurable observation window.

Mitigation Module Design

The mitigation module activates immediately upon receiving an attack alert, before isolation consensus is complete. It implements traffic redirection by temporarily demoting suspected nodes

in the local routing table and redirecting data traffic through alternative parents identified from the trusted node list maintained by the trust computation module. Nodes identified as high-priority for critical IoT service delivery are given preferential alternative routing to maintain throughput within the 10% degradation target.

A quarantine state is implemented for nodes under investigation, during which they are allowed to continue forwarding data (to avoid false-positive service disruption) but are excluded from being selected as preferred parents by neighboring nodes. Emergency communication channels using direct-to-root routing are established for nodes whose only available paths traverse suspected malicious nodes.

Recovery Module Design

Following successful isolation, the recovery module triggers targeted topology reconstruction rather than global DODAG reformation. Nodes that were routing through the isolated node are identified via the existing routing table and directed to initiate local parent re-selection using the trust-enhanced objective function. This localized healing strategy avoids the cascading topology instability associated with global DODAG version incrementation, targeting recovery completion within 30 seconds for networks of up to 100 nodes.

A secure re-authentication sub-module manages requests from nodes seeking to rejoin the network after isolation. Re-entry requests are evaluated against a multi-factor verification procedure including node identity confirmation, behavioral history review, and a supervised re-integration period with enhanced trust monitoring before full routing privileges are restored.

Phase 3: Integration and System Testing

The isolation, mitigation, and recovery modules are integrated with the trust computation engine, the IDS detection module, and the trust-aware parent selection algorithm through standardized inter-module interfaces. Unit testing is conducted for each module independently, followed by integration testing of the complete pipeline from alert generation through recovery completion. System-level testing verifies end-to-end behavior under both individual and combined attack scenarios.

Phase 4: Performance and Security Evaluation

The framework is evaluated against an Evaluation Matrix covering four dimensions: Isolation Effectiveness (consensus accuracy, false isolation rate, isolation latency), Mitigation Performance (throughput retention, service availability, rerouting overhead), Recovery Quality (topology reconstruction time, post-recovery PDR, convergence stability), and Resource Efficiency (additional CPU overhead, memory consumption, and energy cost relative to standard RPL). Evaluation targets include isolation latency below 3 seconds, post-recovery PDR above 98%, less than 10% additional energy overhead, and recovery time within 30 seconds across all evaluated network sizes.

6. Sources for Test Data and Analysis

Test Data Sources

Network simulation data generated by the Contiki-NG operating system and the Cooja network simulator constitutes the primary source of test data. These tools provide controlled and reproducible experimental environments that have been widely validated for RPL security research. Simulation scenarios are constructed across four categories: normal operation, isolated sinkhole attack, isolated selective forwarding attack, and combined simultaneous sinkhole and selective forwarding attack. Each scenario is evaluated at network scales of 20, 50, and 100 nodes under both static and mobility-enabled topologies.

Data collected from each simulation run includes DODAG structure logs capturing parent-child relationships and rank assignments, routing table snapshots before and after isolation events, control message counts (DIO, DAO, DIS) and transmission timestamps, data packet forwarding logs with sequence numbers and delivery outcomes, blacklist propagation records tracking consistency across nodes, and energy consumption metrics per node throughout the simulation.

Additionally, data from the IDS and trust computation modules developed by collaborating team members serves as input to the isolation module. This includes attack alert messages with identified malicious node IDs and confidence scores, trust score time-series for individual nodes, and parent selection decisions from the trust-aware routing module.

Data Collection Process

Data collection is fully automated within the Cooja simulation environment. Each IoT node running the Contiki-NG implementation logs relevant isolation, mitigation, and recovery events using lightweight, timestamped log entries consistent with the resource constraints of the platform. Simulation runs are repeated a minimum of five times per scenario configuration to account for stochastic variation in node behavior and radio link quality models.

For each simulation run, baseline behavior is captured during the initial observation period prior to attack introduction. Attack scenarios are then activated at a predefined simulation time, and all subsequent events — including alert reception, isolation voting, blacklist updates, traffic rerouting, and topology reconstruction — are logged with high-precision timestamps. Recovery completion is marked when the network achieves a stable DODAG with PDR above the 98% target threshold.

Data Analysis Techniques

Simulation log files exported from Cooja are processed using Python scripts that parse event logs, compute performance metrics, and generate statistical summaries. Key analysis procedures include isolation latency measurement from alert reception to blacklist confirmation across all nodes, blacklist consistency evaluation measuring the fraction of nodes with synchronized blacklists within the 3-second target, throughput retention calculation comparing data delivery rates before attack introduction and during the mitigation phase, recovery time measurement from isolation completion to stable post-recovery DODAG formation, and energy overhead comparison between standard RPL baseline and the proposed framework.

Statistical analysis employs descriptive statistics (mean, standard deviation, confidence intervals) across repeated simulation runs. Performance metrics are evaluated against the defined Evaluation Matrix targets to assess whether the framework meets its design objectives. Comparative analysis against standard RPL and selected state-of-the-art approaches provides context for the significance of the achieved improvements.

Validation Strategy

Validation proceeds at three levels. Module-level validation tests each component (isolation, mitigation, recovery) independently against known attack scenarios with predetermined expected outcomes. System-level validation tests the integrated framework against the full attack scenario suite, verifying end-to-end behavior from detection alert to post-recovery stability. Comparative validation benchmarks the framework's performance against standard RPL and the SMTrust and iTRPL approaches cited in the literature, using consistent simulation parameters to ensure valid comparison.

Cross-module validation with the detection and trust computation components developed by collaborating team members verifies that interface contracts are correctly honored, alert messages are correctly interpreted, and coordinated responses produce the expected combined system behavior under all tested attack scenarios.

7. Anticipated Benefits

User and Operational Advantages

IoT network administrators and operators of RPL-based deployments stand to benefit directly from the proposed framework. The most immediate operational benefit is the reduction of service disruption time following a confirmed routing attack. Whereas existing RPL deployments rely on standard DODAG reformation mechanisms that may require 30–90 seconds to converge, the targeted topology reconstruction approach aims to restore full network functionality within 30 seconds, significantly reducing the exposure window for data loss and service unavailability in time-sensitive applications.

The lightweight design of the framework preserves the energy efficiency of IoT devices, extending network operational lifetime compared to solutions employing computationally intensive mitigation strategies. By targeting less than 10% additional energy overhead relative to standard RPL, the framework ensures practical deployability in battery-powered sensor networks where energy conservation is a primary operational constraint.

The secure re-authentication mechanism provides network administrators with confidence that once a malicious node has been identified and isolated, it cannot silently re-enter the network and resume adversarial behavior. This is particularly valuable in deployments where physical replacement of compromised nodes is impractical, as the framework allows for a controlled and monitored re-integration pathway that balances security with operational flexibility.

Contribution to the Body of Knowledge

This research makes several novel contributions to the RPL security literature. Most significantly, it provides the first systematic treatment of the post-detection phase of RPL security management as a distinct and independently addressable research problem, filling a gap repeatedly identified in survey literature but not yet directly addressed.

The distributed consensus-based isolation protocol contributes a formally specified mechanism for consistent, network-wide malicious node removal that does not require centralized control, representing an advance over existing approaches that rely on unilateral local decisions. The targeted topology reconstruction algorithm provides an alternative to global DODAG reformation that is specifically optimized for the security-triggered isolation scenario, contributing a reusable algorithmic component for future RPL resilience research.

The secure re-authentication framework addresses a previously unexamined attack surface — the re-entry of isolated nodes — and provides a lightweight multi-factor verification mechanism that can be adopted or extended by future IoT security research. Together, these contributions advance the practical deployability of trust-aware RPL security mechanisms and provide a validated implementation template for the research community.

The experimental results produced using Contiki-NG and Cooja provide reproducible empirical evidence for the feasibility and performance of the proposed mechanisms under realistic constrained IoT conditions, contributing a replicable experimental baseline for future comparative studies.

8. Scope and Specified Deliverables / Expected Research Outcome

Scope of Research

This research component focuses specifically on the post-detection security management phase of RPL-based IoT network protection, encompassing three interconnected functional areas: distributed malicious node isolation, attack impact mitigation, and network topology recovery. The scope is bounded by the outputs of the IDS and trust computation modules developed by collaborating team members, which provide the detection alerts and trust scores that drive this component's operations.

The framework operates at the RPL routing layer and is designed for deployment in a distributed manner on individual IoT nodes running Contiki-NG. It targets RPL networks of 20 to 100 nodes under both static and mobile topologies, evaluated against sinkhole and selective forwarding attack scenarios individually and in combination. Implementation and evaluation use the Contiki-NG operating system and Cooja network simulator.

Out of Scope

The following elements are explicitly excluded from the scope of this component to maintain focus and feasibility:

- Detection of malicious nodes — this is handled by the IDS component developed by a collaborating team member.
- Trust computation and parent selection algorithms — these are addressed by other components within the project team.
- Application-layer security mechanisms or cryptographic key management.
- Physical hardware deployment on real IoT devices — evaluation is limited to simulation.
- Attack types beyond sinkhole and selective forwarding (e.g., Sybil, wormhole, version number attacks).
- Machine learning or deep learning-based decision mechanisms due to resource constraints.

Specified Deliverables and Expected Research Outcomes

The following deliverables are anticipated from this research component:

1. A formally specified distributed isolation protocol design document, including the consensus mechanism specification, blacklist management protocol, and false-isolation prevention logic.
2. A Contiki-NG implementation of the isolation, mitigation, and recovery modules integrated with the RPL protocol stack and compatible with the project's IDS and trust computation interfaces.
3. Empirical performance evaluation results from Cooja simulations covering isolation latency, mitigation throughput retention, recovery time, and resource overhead across network scales and attack scenarios.
4. A secure re-authentication framework specification and implementation, including the progressive trust rehabilitation protocol and enhanced monitoring procedures for re-integrated nodes.
5. Integration validation results demonstrating correct end-to-end system behavior across the complete detection-to-recovery pipeline.
6. Technical documentation including module design specifications, interface definitions, experimental setup descriptions, and evaluation result reports.

9. Research Constraints

Several constraints bound the conduct of this research, each of which has been considered in the framework design to ensure that the proposed work remains both feasible and academically rigorous.

The primary constraint is the reliance on simulation-based evaluation rather than physical hardware deployment. While Contiki-NG and Cooja provide a well-validated and widely used research platform, simulated environments cannot fully replicate hardware-specific phenomena such as variable radio propagation, manufacturing-level energy variation, and physical node failure modes. This limitation is accepted as appropriate for the preliminary research phase and is consistent with the methodology employed by the broader project team and comparable work in the RPL security literature.

A second constraint concerns the dependency on the IDS and trust computation modules developed by collaborating team members. The isolation, mitigation, and recovery framework's inputs — attack alerts and trust scores — must be provided by these external components, creating an inter-module dependency that requires careful interface management. This dependency is managed through defined inter-module contracts and joint testing sessions, but delays or changes in the collaborating components could affect integration timelines.

Resource constraints inherent to the IoT deployment environment impose a third set of limitations. The prohibition on machine learning and computationally intensive cryptographic operations significantly constrains the design space for isolation consensus and re-authentication mechanisms. This constraint drives the selection of lightweight statistical and rule-based approaches, which may have reduced effectiveness against sophisticated adaptive adversaries compared to learning-based alternatives.

Finally, the evaluation is bounded to sinkhole and selective forwarding attack scenarios, as these are the primary targets of the broader project. The framework's effectiveness against other RPL attack types is not evaluated within this study, which limits the generalizability of the conclusions to the two studied attack categories. Extensions to cover additional attack vectors represent a clear direction for future work.

10. Project Plan or Schedule

The project schedule aligns with the broader team timeline and key academic milestones. The Gantt chart below presents the work breakdown structure and timeline for this individual component, showing the sequencing of major tasks from literature review through final submission.

Task	Oct–Nov 2025	Dec–Jan 2026	Feb–Mar 2026	Apr–Jun 2026
Literature Review & Gap Analysis				
Isolation Protocol Design				
Mitigation Module Design				
Recovery Algorithm Design				
Contiki-NG Implementation				
Unit Testing (Modules)				
Integration Testing (Team)				
Simulation & Evaluation				
Research Paper Writing				
Final Report & Submission				

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